

Chemistry — Assignment 3: Degree of Alcohols, Amines & Halides

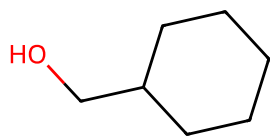
Anshuman Atrey · 150096724029 · BTech CSE 2024-28

Chemistry (Foundational Chemistry) · Semester 5

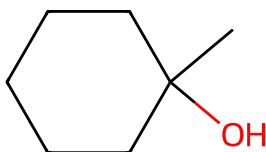
Question. Identify the degree ($1^\circ/2^\circ/3^\circ$) of the given alcohols, amines and halides.

Rule. Classify by the carbon bearing the group ($-\text{OH}$ / $-\text{X}$), or by the nitrogen for amines: count the carbons attached to it $\rightarrow 1^\circ/2^\circ/3^\circ$. (Representative structures matching the set.)

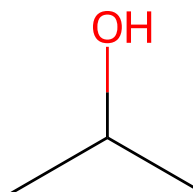
Alcohols



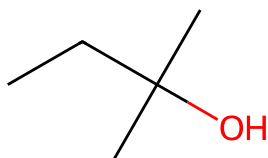
(1) 1° — cyclohexylmethanol



(2) 3° — 1-methylcyclohexan-1-ol

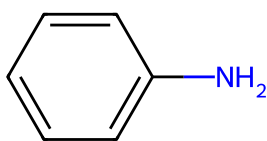


(3) 2° — propan-2-ol

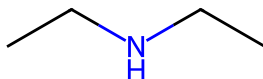


(4) 3° — 2-methylbutan-2-ol

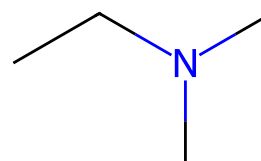
Amines



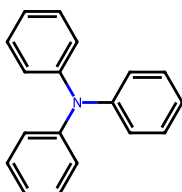
(1) 1° — aniline (phenylamine)



(2) 2° — diethylamine

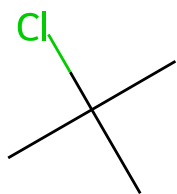


(3) 3° — N,N-dimethylethanamine

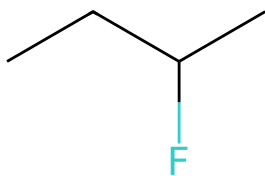


(4) 3° — N,N-diphenylaniline

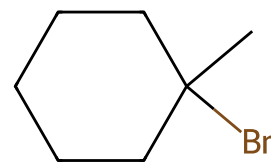
Halides



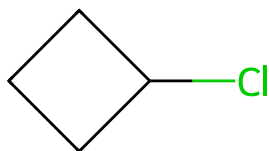
(1) 3° — 2-chloro-2-methylpropane



(2) 2° — 2-fluorobutane



(3) 3° — 1-bromo-1-methylcyclohexane



(4) 2° — chlorocyclobutane